

MT-KD: Multi-Teacher Knowledge Distillation for Privacy-Aware Unlearning in IIoT Intrusion Detection Systems

Shigen Shen[✉], Senior Member, IEEE, Jiashuo Niu[✉], Yizhou Shen[✉], Jingnan Dong[✉], Member, IEEE, Wenlong Ke[✉], Tian Wang[✉], Senior Member, IEEE, and Ruidong Li[✉], Senior Member, IEEE

Abstract—The growth of Industrial Internet of Things (IIoT) systems creates a direct conflict for intrusion detection systems: while essential for security, their inability to selectively forget specific attack patterns hinders adaptability and impedes compliance with privacy regulations like General Data Protection Regulation. To resolve this conflict, we introduce the Multi-Teacher Knowledge Distillation (MT-KD) unlearning framework, designed to erase knowledge of specified attack classes without the prohibitive cost of full model retraining. Our approach employs deep convolutional neural networks, multi-teacher guidance, and gradient-based noise injection to train a student model that forgets targeted attacks while retaining high accuracy on remaining classes. We validate this by converting network traffic from the CIC-IDS2017 and USTC-TFC2016 benchmark datasets into grayscale images for extensive experiments. The results demonstrate that our MT-KD framework achieves complete forgetting of targeted classes, verified by Membership Inference Attack resistance, while maintaining over 95% accuracy on retained classes. By outperforming existing unlearning baselines in both effectiveness and efficiency, our work delivers a practical and scalable solution for privacy-preserving intrusion detection in real-world IIoT scenarios. The source code is publicly available at <https://github.com/NJS-HZ/MT-KD>

Index Terms—Machine unlearning, knowledge distillation, Industrial Internet of Things, intrusion detection system, privacy.

I. INTRODUCTION

INDUSTRIAL Internet of Things (IIoT) technologies are changing manufacturing, energy, and critical infrastructure

systems [1]. They do this by making it possible to get data in real time and use smart automation [2]. But, all these new connections and complex systems also lead to more advanced cyber threats that target industrial control systems, edge devices, and communication protocols [3], [4]. As a result, Intrusion Detection Systems (IDS) have become indispensable for safeguarding IIoT intrusion detection environments against diverse and evolving attacks [5], [6]. Nevertheless, they often retain long-term memory of obsolete or sensitive threat patterns, which can pose significant risks to both privacy compliance and the system's adaptability to emerging attack vectors. In IIoT settings—where operational continuity, data protection, and regulatory alignment are paramount—there is a growing demand for mechanisms that can selectively remove previously learned information from deployed detection models [7].

Regulations such as the EU's General Data Protection Regulation [8] and California Consumer Privacy Act [9] grant individuals the *right to be forgotten* and to request deletion of personal data [10]. In practice, compliance extends beyond erasing raw records: if models were trained on those data, their parameters may encode them, and organizations may need to remove that influence from deployed models as well [11], [12]. For IIoT-based learning systems in mission-critical and data-sensitive settings, it needs a practical requirement to support privacy-compliant unlearning that selectively remove the effect of specified samples or classes from a trained model without retraining from scratch [13].

Machine unlearning has moved beyond computer vision to natural language processing and recommender systems [14], with techniques ranging from influence-function-guided parameter edits and Newton-style update removal to partitioned training schemes with certified deletion guarantees. By contrast, applications to IIoT cybersecurity remain sparse [15], [16]. Much of the literature still chases stronger intrusion detection—robustness [17], higher detection accuracy, or incremental learners under concept drift—while the more prosaic yet crucial task of systematically scrubbing models of outdated or sensitive attack patterns is rarely addressed [18].

Given the practical limits of IIoT—such as limited compute at terminals, mixed protocols [19], and tight latency needs—retraining a model from scratch to forget one or more attack types is not practical. So recent machine unlearning work [20] has proposed fast and selective erase methods

Received 20 October 2025; revised 8 February 2026; accepted 12 February 2026. Date of publication 18 February 2026; date of current version 27 February 2026. This work was supported in part by the Humanities and Social Sciences Planning Foundation of Ministry of Education of China under Grant 24YJAZH123, and Huzhou Science and Technology Planning Foundation of China under Grant 2023GZ04. The associate editor coordinating the review of this article and approving it for publication was J. Kang. (Corresponding author: Yizhou Shen.)

Shigen Shen, Jiashuo Niu, Jingnan Dong, and Wenlong Ke are with the School of Information Engineering, Huzhou University, Huzhou 313000, China (e-mail: shigens@zjhu.edu.cn; 2024388229@stu.zjhu.edu.cn; dongjingnan@zjhu.edu.cn; wlke@zjhu.edu.cn).

Yizhou Shen is with the School of Computing, Newcastle University, NE1 7RU Newcastle upon Tyne, U.K. (e-mail: Y.Shen34@newcastle.ac.uk).

Tian Wang is with the Institute of Artificial Intelligence and Future Networks, Beijing Normal University, Zhuhai 519087, China, also with the College of Computer and Data Science, Fuzhou University, Fuzhou 350108, China, and also with the College of Computer Science and Mathematics, Fujian University of Technology, Fuzhou 350118, China (e-mail: tianwang@bnu.edu.cn).

Ruidong Li is with the Institute of Science and Engineering, Kanazawa University, Kakuma, Kanazawa 920-1192, Japan (e-mail: lrd@se.kanazawa-u.ac.jp).

Digital Object Identifier 10.1109/TCCN.2026.3665917

that keep compute cost low. These include influence-function updates, small local weight tweaks, and knowledge distillation with a teacher model. Even with these advances, balancing unlearning strength with overall model stability and detection performance remains a key challenge [21], [22].

To address these challenges, we propose a novel machine learning forgetting framework specifically designed for IIoT intrusion detection scenarios. The proposed method—Multi-Teacher Knowledge Distillation (MT-KD)—combines the newly developed UnResNet9 model, tailored for learning elimination tasks, with a multi-teacher selective forgetting strategy. This framework removes knowledge of specific attack classes from pretrained detection models while keeping performance on the remaining categories. Its core is a student model trained with guidance from multiple teachers, using both noise-based forgetting and distillation with limited data. This design enables precise, controllable forgetting without retraining the whole model from scratch.

We present a full unlearning pipeline that identifies the samples or classes to forget, generates error-maximizing noise to push the model away from them, and uses a multi-stage alternating training schedule to erase and then stabilize the model. We convert raw network traffic into 28×28 grayscale session images with the USTC-TK2016 traffic-to-image toolkit [23], and evaluate on two public benchmarks: CIC-IDS2017¹ and USTC-TFC2016.² CIC-IDS2017 covers enterprise attacks including DoS, DDoS, brute force, web attacks, and botnets, while USTC-TFC2016 contains real malicious traffic from families such as Neris, Virut, and Geodo. Results across both datasets show good generalization and robustness to different network settings and attack types: our method reliably forgets targeted attack categories while keeping high detection accuracy on benign traffic and retained classes, outperforming full retraining and prior unlearning baselines.

In summary, our contributions are three-fold:

- We propose a lightweight residual network, UnResNet9, built for IIoT intrusion detection and aimed at the inter-class learning loss issue. Unlike standard ResNet variants, UnResNet9 adds adaptive feature layers and lightweight regularization so the model can prune unneeded knowledge while keeping detection performance on the attack categories that remain.
- Our multi-teacher knowledge distillation (MT-KD) setup uses two teachers. The forgetting teacher is obtained by noise training the original pretrained model with error-maximizing perturbations aimed at the classes to forget, which yields a strong forgetting signal. The retention teacher is trained only on the retained dataset and provides clean guidance for the remaining classes. The student learns useful knowledge from the retention teacher and follows the forgetting cues from the forgetting teacher by adding loss terms on noise-perturbed samples and suppressing the logits of the target classes during training. This design enables precise class-level forgetting without retraining from scratch, reduces compute and

latency, and keeps detection performance strong on the retained categories.

- We conduct extensive evaluations on two benchmark datasets: CIC-IDS2017 and USTC-TFC2016. Both datasets are preprocessed into grayscale images using the USTC-TK2016 traffic-to-image conversion toolkit [23], enabling CNN-based classification. Experimental results demonstrate that our MT-KD framework achieves complete forgetting of the targeted attack category, which has been validated as resistance to membership inference attacks. Simultaneously, it maintains over 95% accuracy in the retained categories, thereby surpassing existing forgetting baselines in both effectiveness and efficiency.

The remainder of this paper is organized as follows. Section II reviews related work on machine unlearning, IIoT intrusion detection, and knowledge distillation. Section III presents the system model and data configuration. Section IV details the proposed MT-KD unlearning framework, including data preprocessing, the UnResNet9 architecture, and the multi-teacher knowledge distillation mechanism. Section V presents comprehensive experimental results and analysis on two benchmark datasets. Finally, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

A. Machine Unlearning

Recent years have witnessed growing interest in machine unlearning as a critical technique for protecting user privacy and enforcing the *right to be forgotten* in machine learning systems. Hu et al. [24] exposed privacy vulnerabilities during unlearning by proposing inversion attacks that exploit both original and unlearned models. Sun et al. [25] addressed unlearning challenges in GANs through a cascaded approach preserving latent space continuity. Tarun et al. [26] developed a fast framework leveraging error-maximizing noise and weight manipulation for class unlearning without full data access. Wang et al. [27] proposed Federated Unlearning (FedU) for federated scenarios, balancing efficacy and utility through user-side influence approximation. Chundawat et al. [28] introduced zero-shot unlearning with error-based noise optimization and the Anamnesis Index metric for quality evaluation.

Collectively, recent studies have made steady progress on machine unlearning across model types and setups, mainly classifiers and generative models, with a focus on speed, accuracy, and privacy. Yet several open issues remain: cutting the drop in performance during unlearning, making the unlearning pipeline secure from start to finish, and building general frameworks that comply with strict regulations [29], [30]. These gaps show up across datasets and architectures and call for clearer evaluation and deployment practices.

B. Intrusion Detection in IIoT Environments

IIoT networks face considerable challenges due to heterogeneous and severely imbalanced data collected from diverse terminal devices, where data distributions are often

¹Available at: <https://www.unb.ca/cic/datasets/ids-2017.html>

²Available at: <https://github.com/yungshenglu/USTC-TFC2016>

non-independent and identically distributed with under-represented attack types. To address these challenges, Shan et al. [31] proposed Clustered Federated Learning that leverages dynamic focal loss and model clustering to enhance robustness against poisoning attacks. Hu et al. [32] conducted a comprehensive survey on IIoT intelligence, proposing a five-layer hierarchical architecture and examining enabling technologies to guide future research. Yu et al. [33] introduced a deep reinforcement learning-based open-set IDS that effectively identifies both known and unseen attack types, addressing zero-day threat detection challenges. Zhang et al. [34] proposed an unsupervised fault detection framework for multi-condition industrial signals, supporting intelligent manufacturing and predictive maintenance in IIoT systems.

Overall, recent work points to a trend of combining federated learning, reinforcement learning, and machine unlearning to build IIoT systems that are scalable and robust to advanced attacks, and privacy-aware, addressing the diverse and fast-changing nature of IIoT [35], [36].

C. Knowledge Distillation

In recent years, knowledge distillation has proven effective for model compression and transfer [37] and is now used in machine unlearning to selectively remove class- or instance-level knowledge. Compared with retraining from scratch, distillation-based methods cut compute and time, making them a good fit for efficient unlearning in deep models [38]. In particular, MT-KD improves student robustness and capacity by using guidance from multiple teachers. Building on this, we propose MT-KD-Unlearning, which trains the student through collaborative learning on retained data together with substitute noise samples. This achieves precise forgetting of malicious attack categories while maintaining high performance on the retained classes.

D. Our Method

The above analysis shows that distillation-based forgetting methods are promising for selective forgetting, but they also reveal a key drawback: strong reliance on a single teacher model. This limitation makes it hard to control, in a detailed way, what to keep and what to remove in complex, multi-class cybersecurity settings.

To address these limits, we propose MT-KD Forgetting Learning for IIoT intrusion detection. It uses two teachers with complementary roles: a forgetting teacher, obtained by noise training the original model to emphasize what should be dropped, and a retention teacher trained only on the retained data to provide clean guidance. The student alternates between two training phases. In one phase, it distills knowledge from the retention teacher on retained samples. In the other, it learns to suppress responses to the target classes using noise-perturbed inputs guided by the forgetting teacher. The noise strength is kept modest to trigger forgetting without destabilizing training, and the alternation is kept short to maintain stability. The balance between the two phases can be adjusted to trade off forgetting strength and retained accuracy. The

approach works with common backbones, leaves the inference path unchanged, and does not require access to the original data that must be removed, which suits privacy-sensitive deployments. In practice, it adds only a small training-time overhead while keeping test-time cost unchanged, and it preserves detection on retained classes while removing knowledge of the targets.

The specific design of MT-KD is particularly well-suited for IIoT intrusion detection environments. First, unlike standard distillation-based unlearning methods that rely on expensive parameter modification, our noise tensor optimization significantly reduces computational overhead, enabling practical deployment on IIoT edge devices. Second, the dual-teacher architecture with strategic MaxPooling bottlenecks enables class-specific forgetting boundaries, allowing fine-grained control over which attack classes are forgotten while minimizing collateral damage to retained threat detection. Standard methods lack this precision and typically resort to coarser global forgetting strategies unsuitable for multi-class cybersecurity scenarios. Third, our lightweight UnResNet9 backbone is optimized for 28×28 grayscale network traffic images rather than arbitrary image sizes, and its design captures the periodic byte patterns of industrial protocols such as MODBUS and DNP3. In contrast, directly adapting general-purpose CNNs to IIoT would ignore these protocol-specific structures. These design choices collectively address the unique constraints of IIoT systems—resource limitations, privacy requirements, and the need for selective, multi-class unlearning—making MT-KD substantially more practical and effective than direct adaptations of existing methods.

III. PROPOSED MODEL

A. System Model

In the IIoT model, edge devices send their network traffic to a central server for analysis. This stream contains both normal communications and malicious traffic injected by attackers. To enhance security and protect sensitive operational data, we introduce a noise-based machine forgetting algorithm to eliminate harmful samples. This mechanism aims to mitigate malware impacts and erase attack traces during the learning process. Through continuous iterative forgetting until malicious knowledge is completely purged, we ultimately construct a sophisticated, privacy-focused model resilient against industrial cyber threats.

In the framework of the unlearning algorithm depicted in Fig. 1, the data of each user exhibits randomness and uncertainty, and the attack methods employed by attackers are also unpredictable. The unlearning algorithm continuously learns and sifts out malicious data for the purpose of forgetting it. After being processed by the server side, the relevant data is uploaded to the final model repository. The model we have generated is a high-performance one that has screened out malicious data. In this way, it can protect users' network security and privacy, and enhance the overall security level.

At the practical level, forgetting learning lets security models quickly dampen the impact of specific attack cases or tainted samples and recover detection quality without rebuilding the whole model. Placed on edge intelligence nodes,

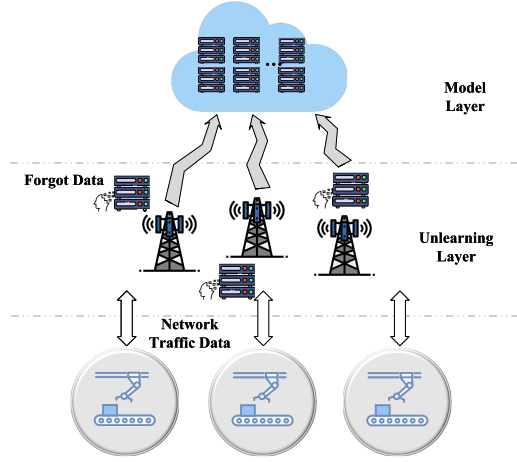


Fig. 1. Layered architecture of MT-KD unlearning in IIoT intrusion detection environments.

an IDS can make small, local parameter updates to forget harmful signals while still learning normal traffic patterns. This avoids the delay and bandwidth cost of round-trip cloud updates and keeps the system capable of self-defense even when offline or under tight resource limits. When appropriate, it can also be paired with federated learning so devices coordinate forgetting steps without sending raw data, helping protect a shared model with little communication. Overall, the lightweight forgetting path improves real-time response in distributed settings, reduces reliance on the cloud, and strengthens resilience against adversarial interference and data-protection risks.

We are currently exploring practical ways to remove malicious data from networked systems while keeping retained data intact, especially when hostile traffic attempts unauthorized access to network resources and users' private information. In real deployments, malware reaches privacy-sensitive nodes through common paths such as phishing, software flaws, and lateral movement from compromised hosts; zero-day attacks are among the most common and hardest to handle. To deal with these problems, we propose a machine-learning unlearning framework that supports selective forgetting of malicious traces while preserving benign knowledge and user privacy. The aim is a cleaner, safer model that resists malware and better protects data integrity and privacy.

B. Data Configuration

We consider a supervised learning dataset $D = \{(x_i, y_i)\}_{i=1}^N$, where each x_i denotes an input sample and $y_i \in \{1, 2, \dots, K\}$ represents its corresponding class label, with K denoting the total number of distinct classes.

We define the subset of classes that the model is intended to forget $\mathcal{F} \subseteq \{1, 2, \dots, K\}$, referred to as the *forget class set*. The corresponding subset of data is defined as

$$\mathcal{D}_{\mathcal{F}} = \{(x_i, y_i) \in D \mid y_i \in \mathcal{F}\}. \quad (1)$$

where $\mathcal{D}_{\mathcal{F}}$ denotes the forget data subset, and $y_i \in \mathcal{F}$ indicates that the label of sample (x_i, y_i) belongs to the forget class set.

Similarly, the set of classes that the model is intended to retain is defined as $\mathcal{R} = \{1, 2, \dots, K\} \setminus \mathcal{F}$, referred to as the *retain class set*. The associated data subset is:

$$\mathcal{D}_{\mathcal{R}} = \{(x_i, y_i) \in D \mid y_i \in \mathcal{R}\}. \quad (2)$$

where $\mathcal{D}_{\mathcal{R}}$ denotes the retain data subset, and $y_i \in \mathcal{R}$ indicates that the label of sample (x_i, y_i) belongs to the retain class set.

This partitioning lays the foundation for selective unlearning algorithms, enabling the model to effectively remove information related to specific classes while preserving knowledge deemed important. The complete dataset D can be decomposed into two mutually exclusive subsets:

$$D = \mathcal{D}_{\mathcal{F}} \cup \mathcal{D}_{\mathcal{R}}, \quad \mathcal{D}_{\mathcal{F}} \cap \mathcal{D}_{\mathcal{R}} = \emptyset. \quad (3)$$

To evaluate the effect of unlearning, we construct two models for comparison: the *original model*, which is trained on the full dataset D , and the *unlearning model*, which is trained solely on the retained dataset $\mathcal{D}_{\mathcal{R}}$, with all forget data $\mathcal{D}_{\mathcal{F}}$ excluded.

IV. MULTI-TEACHER KNOWLEDGE DISTILLATION UNLEARNING MODEL

A. Data Preprocessing

To use deep learning for IIoT intrusion detection, we first convert raw IIoT traffic into forms that models can learn. We parse packet captures into flows, extract session-level features, and pack them into fixed-size tensors suitable for convolutional models. In this work we use the public CIC-IDS2017 and USTC-TFC2016 datasets, which cover a wide range of benign and attack traffic and show clear temporal patterns and diverse communication behavior. We apply a consistent preprocessing pipeline across both datasets with flow slicing, per-feature normalization, and stratified train, validation, and test splits. This setup makes the inputs stable for training and helps the models focus on the real differences between benign and malicious traffic.

Both CIC-IDS2017 and USTC-TFC2016 cover the common threats and traffic patterns seen in IIoT intrusion detection. Attacks such as DoS, PortScan, and Botnet target key assets in industrial control systems and are well represented in these datasets. The traffic records show typical exchanges between industrial endpoints and servers, providing useful training material for IIoT-focused detectors. Using these datasets helps models adapt to real deployments and remain robust under practical conditions.

The dataset is originally provided in the form of raw packet capture (PCAP) files, which record continuous streams of network packets. Since such raw data is not directly suitable for deep convolutional models, we begin by segmenting the continuous traffic trace $\mathcal{T}$ into fixed-duration session-based flow windows using a slicing function as

$$\phi(\mathcal{T}; \tau) = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N\}, \quad (4)$$

where $\mathcal{T}$ denotes the continuous traffic trace, τ is the fixed time duration for session segmentation, $\mathbf{s}_i$ represents the i -th flow segment, and N is the total number of extracted sessions.

Each flow segment $\mathbf{s}_i$ is a time-aligned session of length τ between two endpoints, covering the protocol exchanges and

basic behavior. We process these segments with the USTC-TK2016 toolkit and extract a session feature vector $\mathbf{f}_i \in \mathbb{R}^d$, where d is the number of features. The vector records simple statistics and protocol fields, such as packet inter-arrival times, byte counts, flow duration, and TCP flag ratios.

To adapt these features for CNNs, each vector is normalized, reshaped, and mapped into a grayscale image tensor through a transformation function ψ as

$$\psi(\mathbf{f}_i) \in \mathbb{R}^{C \times H \times W}. \quad (5)$$

This conversion preserves spatial correlations within the features and allows convolutional filters to detect complex flow patterns similarly to textures in natural images.

To improve model robustness and enable the detection of zero-day and previously unseen attacks, we further augment the dataset with synthetic samples $\tilde{\mathbf{f}}_j$ generated through three complementary strategies. First, we apply feature-level synthesis by perturbing and recombining existing attack vectors to simulate polymorphic behavior, which is represented by

$$\tilde{\mathbf{f}} = \mathbf{f}^{(a)} + \epsilon \odot (\mathbf{f}^{(b)} - \mathbf{f}^{(c)}), \quad \epsilon \sim \mathcal{U}(0, 1)^d. \quad (6)$$

where $\mathbf{f}^{(a)}, \mathbf{f}^{(b)}, \mathbf{f}^{(c)} \in \mathbb{R}^d$ denote three different known attack feature vectors, $\epsilon \in \mathbb{R}^d$ is a random vector with each element independently sampled from the uniform distribution $\mathcal{U}(0, 1)$, and $\odot$ denotes element-wise (Hadamard) product.

Next, we generate adversarial examples by leveraging the gradients of a pretrained model $\mathcal{M}$ to modify known features, pushing them towards the model's decision boundaries, which is represented by

$$\tilde{\mathbf{f}} = \mathbf{f} + \eta \cdot \text{sign}(\nabla_{\mathbf{f}} \mathcal{P}(\mathcal{M}(\mathbf{f}), y)), \quad (7)$$

where $\mathbf{f} \in \mathbb{R}^d$ denotes the original input feature vector, $\tilde{\mathbf{f}}$ is the perturbed adversarial example, $\eta > 0$ controls the magnitude of the perturbation, $\text{sign}(\cdot)$ is the element-wise sign function, $\nabla_{\mathbf{f}} \mathcal{P}(\cdot)$ represents the gradient of the loss with respect to the input, indicating the sensitivity of the model output to changes in each input dimension, $\mathcal{M}(\mathbf{f})$ denotes the model prediction given $\mathbf{f}$, and y is the true label. This method effectively enhances the model's robustness against adversarial evasion attacks.

Finally, we train generative models $G(\mathbf{z})$, such as GANs [39] or Variational Autoencoders (VAEs) [40], on known attack samples to learn their underlying distribution $p_{\text{attack}}(\mathbf{f})$, and generate new synthetic samples by sampling latent variables $\mathbf{z}$ from a multivariate standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$, which is represented by

$$\tilde{\mathbf{f}} = G(\mathbf{z}), \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (8)$$

All synthetic features are assigned a unified label y_{unknown} to indicate their unknown or novel nature, and are transformed into grayscale images $\mathbf{I} = \psi(\tilde{\mathbf{f}})$ using the same preprocessing pipeline. The final augmented training dataset is constructed as

$$\mathcal{D}_{\text{train}}^{\text{aug}} = \mathcal{D}_{\mathcal{R}} \cup \{(\psi(\tilde{\mathbf{f}}_j), y_{\text{unknown}})\}_{j=1}^M. \quad (9)$$

Here $\mathcal{D}_{\text{train}}^{\text{aug}}$ denotes the augmented training dataset composed of the retained data subset $\mathcal{D}_{\mathcal{R}}$ and M synthetic samples. Recall that $\mathcal{D}_{\mathcal{R}}$ is the retained dataset excluding all forget

data $\mathcal{D}_{\mathcal{F}}$, as defined in Section III-B. This data augmentation process expands the training distribution and decision boundaries of retained classes, thereby improving the model's generalization ability and enabling more effective detection of previously unseen and evolving attack variants.

B. Convolutional Model UnResNet9

To handle grayscale traffic images, we use a small, efficient CNN called UnResNet9, a customized version of ResNet9. It keeps the key feature layers of ResNet9 and adds a few simple changes so it works well with multi-teacher knowledge distillation and selective forgetting. In our MT-KD unlearning framework, UnResNet9 serves as the backbone, learning useful features from network traffic while enabling fast, class-specific forgetting.

Given an input tensor $\mathbf{I} \in \mathbb{R}^{C \times H \times W}$, where $C = 1$ for grayscale images, $H = W = 28$ for our dataset, the model extracts hierarchical and discriminative feature embeddings from flow-level image representations through a sequence of convolution operations, batch normalization, nonlinear activation functions, and residual connections. Specifically, each convolutional layer consists of a 2D convolution operation $\text{Conv}_k(\cdot)$ followed by batch normalization $\text{BN}_k(\cdot)$, formally expressed as

$$\text{ConvBN}_k(\mathbf{U}) = \text{BN}_k(\text{Conv}_k(\mathbf{U})), \quad (10)$$

where $\mathbf{U}$ denotes the input feature tensor, and the subscript k denotes the layer index. Batch normalization stabilizes training after the convolution operation.

To facilitate deeper feature learning and mitigate gradient vanishing issues, we incorporate residual blocks $\mathcal{R}(\cdot)$ defined by

$$\mathcal{R}(\mathbf{U}) = \sigma(\text{ConvBN}_2(\sigma(\text{ConvBN}_1(\mathbf{U}))) + \mathbf{U}), \quad (11)$$

where $\sigma(\cdot)$ denotes the ReLU activation function, the subscripts 1 and 2 denote the two convolutional layers within the residual block, and the residual connection enables cross-layer information flow by adding the input $\mathbf{U}$ to the output of two consecutive convolution-batch normalization-activation operations.

The forward propagation of the network computes intermediate feature maps $\mathbf{Z}_1$ through $\mathbf{Z}_{10}$ sequentially as

$$\begin{aligned} \mathbf{Z}_1 &= \sigma(\text{ConvBN}_1(\mathbf{I})), \\ \mathbf{Z}_2 &= \sigma(\text{ConvBN}_2(\mathbf{Z}_1)), \\ \mathbf{Z}_3 &= \text{MaxPool}(\mathbf{Z}_2), \\ \mathbf{Z}_4 &= \mathcal{R}(\mathbf{Z}_3), \\ \mathbf{Z}_5 &= \sigma(\text{ConvBN}_3(\mathbf{Z}_4)), \\ \mathbf{Z}_6 &= \text{MaxPool}(\mathbf{Z}_5), \\ \mathbf{Z}_7 &= \sigma(\text{ConvBN}_4(\mathbf{Z}_6)), \\ \mathbf{Z}_8 &= \text{MaxPool}(\mathbf{Z}_7), \\ \mathbf{Z}_9 &= \mathcal{R}(\mathbf{Z}_8), \\ \mathbf{Z}_{10} &= \text{MaxPool}(\mathbf{Z}_9), \end{aligned} \quad (12)$$

where $\text{MaxPool}(\cdot)$ denotes the max pooling operation with kernel size 2×2 .

Finally, the output feature map $\mathbf{Z}_{10}$ is flattened into a feature vector $\mathbf{h} \in \mathbb{R}^{d_h}$ and passed through a fully connected linear layer with weight matrix $\mathbf{W} \in \mathbb{R}^{K \times d_h}$ and bias vector $\mathbf{b} \in \mathbb{R}^K$ to produce the classification logits as

$$\mathbf{o} = \mathbf{W}\mathbf{h} + \mathbf{b}, \quad (13)$$

where $\mathbf{h}$ is the flattened feature vector, d_h is the dimension of the flattened features, K is the number of output classes (as defined in Section III-B), and $\mathbf{o} \in \mathbb{R}^K$ is the output logit vector corresponding to predicted class scores. This architecture balances computational efficiency and expressive power, making it well-suited for image-based intrusion detection tasks.

UnResNet9 distinguishes itself through strategic design choices that balance computational efficiency with unlearning effectiveness. We design UnResNet9 as a lightweight backbone specifically tailored for multi-teacher knowledge distillation and selective forgetting in IIoT intrusion detection. The architecture employs only two residual blocks at 128 and 256 channels, reducing parameters by approximately 60% compared to ResNet18, enabling efficient deployment on resource-constrained IIoT edge devices. Importantly, UnResNet9's architecture is specifically optimized for 28×28 grayscale traffic images rather than standard larger input sizes, avoiding redundant feature extraction layers. Four strategically placed MaxPool2d operations create information bottlenecks that serve as isolation boundaries during forgetting, preventing unintended information leakage between forgotten and retained classes. This lightweight design makes UnResNet9 more sensitive and responsive to noise-based forgetting strategies, facilitating more effective class-specific unlearning. Additionally, UnResNet9 returns intermediate activations $\{\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_5, \mathbf{Z}_7\}$ for multi-level supervision during knowledge distillation, enabling complementary guidance at different abstraction levels.

C. Unlearning Model

Our objective in selective forgetting is to eliminate a model's predictive capability on a specific subset of data $\mathcal{D}_{\mathcal{F}}$, which contains one or more target classes intended for removal. Rather than retraining the entire model or directly modifying model parameters—approaches that are computationally expensive, risk catastrophic forgetting, and are impractical in real-world IIoT deployments—we adopt a more efficient noise-based optimization approach. This method learns an additive noise tensor $\mathbf{N}$ that can be applied to forget dataset samples without altering the model's frozen parameters. This strategy offers significant advantages: it maintains the model's core architecture and performance on retain data $\mathcal{D}_{\mathcal{R}}$, reduces computational cost by optimizing only a fixed-size tensor rather than millions of model parameters, and naturally enables multi-teacher frameworks by generating diverse noise configurations.

When applied to inputs from the forget dataset $\mathcal{D}_{\mathcal{F}}$, this noise causes the model to misclassify or reduce its confidence on forgotten classes. Formally, the optimization problem for learning the optimal noise is defined as

$$\mathbf{N}^* = \arg \min_{\mathbf{N}} [-\mathcal{L}(\mathcal{M}(\mathbf{X} + \mathbf{N}), \mathbf{y}) + \lambda \|\mathbf{N}\|_F^2], \quad (14)$$

where $\mathbf{N} \in \mathbb{R}^{C \times H \times W}$ is the learnable additive noise tensor applied to all samples in $\mathcal{D}_{\mathcal{F}}$, $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$ is a batch of input samples from $\mathcal{D}_{\mathcal{F}}$ with batch size B , $\mathbf{y} \in \mathbb{R}^B$ is the corresponding label vector, $\mathcal{M}(\cdot)$ denotes the pretrained model with frozen weights, $\mathcal{L}(\cdot, \cdot)$ is the cross-entropy loss function, $\lambda > 0$ is the regularization coefficient controlling noise magnitude, and $\|\cdot\|_F$ denotes the Frobenius norm. The negative loss term encourages the model to produce incorrect predictions on noisy inputs, while the regularization term prevents excessive perturbation. The resulting model, referred to as the first teacher model T_1 , has undergone forgetting via additive noise and exhibits degraded performance on $\mathcal{D}_{\mathcal{F}}$ while maintaining general performance on $\mathcal{D}_{\mathcal{R}}$.

To refine the unlearning process and reduce collateral damage to $\mathcal{D}_{\mathcal{R}}$, we employ a knowledge distillation framework. A second teacher model T_2 is constructed by fine-tuning the original model exclusively on the retained dataset $\mathcal{D}_{\mathcal{R}}$, defined as

$$T_2 = \arg \min_{\mathcal{M}} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\mathcal{R}}} \mathcal{L}(\mathcal{M}(\mathbf{x}), \mathbf{y}), \quad (15)$$

where T_2 denotes the second teacher model obtained by fine-tuning the original model exclusively on the retained dataset $\mathcal{D}_{\mathcal{R}}$. This produces a clean reference model that retains full knowledge of $\mathcal{D}_{\mathcal{R}}$ with no exposure to $\mathcal{D}_{\mathcal{F}}$.

Finally, we train a student model $\mathcal{S}$ that simultaneously learns from both teacher models using a multi-teacher knowledge distillation loss. For an input $\mathbf{x}$, the student attempts to match the softened output distributions of T_1 and T_2 :

$$\begin{aligned} \mathcal{L}_{\text{distill}} = & \alpha \cdot \text{KL}(\mathcal{S}(\mathbf{x}) \parallel T_1(\mathbf{x})) \\ & + (1 - \alpha) \cdot \text{KL}(\mathcal{S}(\mathbf{x}) \parallel T_2(\mathbf{x})), \end{aligned} \quad (16)$$

where $\text{KL}(\cdot \parallel \cdot)$ is the Kullback–Leibler divergence computed on temperature-scaled softmax outputs, $\alpha \in [0, 1]$ balances the influence between forgetting strength (T_1) and performance preservation (T_2), and $\mathcal{S}$ denotes the student model. The student $\mathcal{S}$ is trained only on $\mathcal{D}_{\mathcal{R}}$, ensuring it mimics the desired behavior: poor performance on $\mathcal{D}_{\mathcal{F}}$ while maintaining accuracy on the retained classes.

In summary, our framework provides a flexible and modular approach to machine unlearning, combining noise injection with knowledge transfer to selectively erase learned knowledge while retaining generalization capability. Moreover, it avoids retraining from scratch and ensures compliance with privacy or removal requests in practical systems.

D. Multi-Teacher Knowledge Distillation Model

To effectively integrate complementary knowledge, the student model $\mathcal{S}$ is trained under the joint guidance of two distinct teachers. Teacher T_1 is obtained by applying the learned noise $\mathbf{N}^*$ to the original model's predictions on forget data $\mathcal{D}_{\mathcal{F}}$, thereby achieving degraded performance on forgotten classes. Teacher T_2 is trained solely on the retained data $\mathcal{D}_{\mathcal{R}}$ to preserve important knowledge of retained classes.

Let $\mathcal{L}_{\text{KD}}^{(T_i)}$ be the distillation loss for teacher T_i , which is defined as

$$\mathcal{L}_{\text{KD}}^{(T_i)} = \text{KL} \left(\text{softmax} \left(\frac{f_{\mathcal{S}}(\mathbf{x})}{\tau} \right) \parallel \text{softmax} \left(\frac{f_{T_i}(\mathbf{x})}{\tau} \right) \right), \quad (17)$$

where $\text{KL}(\cdot\|\cdot)$ is the Kullback–Leibler divergence measuring the difference between the student’s and teacher’s output distributions, $f_{T_i}(x)$ and $f_S(x)$ are the logits (pre-softmax outputs) for teacher T_i and student S respectively, $\text{softmax}(\cdot)$ is the softmax function, and $\tau > 0$ is the temperature parameter that controls the smoothness of the output distributions. A higher temperature creates softer probability distributions, which facilitates the transfer of dark knowledge from teachers to student.

Let $\mathcal{L}_{\text{student}}$ be the overall training objective of the student model, which integrates a standard supervised cross-entropy loss on the retained data with a weighted sum of the distillation losses from both teachers:

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{CE}}(f_S(x), y) + \gamma \left[\alpha \mathcal{L}_{\text{KD}}^{(T_1)} + (1 - \alpha) \mathcal{L}_{\text{KD}}^{(T_2)} \right], \quad (18)$$

where $\mathcal{L}_{\text{CE}}(\cdot, \cdot)$ is the cross-entropy loss calculated between the student’s predictions and the ground truth labels y on the retained dataset $\mathcal{D}_{\mathcal{R}}$, $\gamma > 0$ is a hyperparameter balancing the contribution of the distillation losses against the supervised loss, and $\alpha \in [0, 1]$ controls the relative influence of the forgetting teacher T_1 and the clean teacher T_2 .

This multi-teacher distillation strategy enables the student model to effectively suppress knowledge of the forgotten classes by learning from T_1 , while maintaining strong performance on retained classes by learning from T_2 . The rationale is that T_1 provides explicit forgetting guidance through noise optimization, while T_2 trained only on retain data ensures performance preservation. By distilling from both teachers simultaneously, the student model learns two complementary objectives: active suppression of forget class knowledge from T_1 and robust retention of performance from T_2 . This dual-teacher approach transforms unlearning from passive data-driven forgetting into an actively supervised process with measurable forgetting signals. The following experiments sufficiently demonstrate that dual-teacher distillation is necessary for achieving both effective unlearning and strong retention performance.

To show our MT-KD framework balancing machine unlearning and performance preservation, we present Algorithm 1 including three phases: Data preprocessing, unlearning model construction, and student model generation and training.

V. EXPERIMENT

We assess the proposed approach in a multi-teacher unlearning setting across several neural architectures, with emphasis on single- and multi-class forgetting for IIoT intrusion detection. Experiments are conducted on the CIC-IDS2017 and USTC-TFC2016 dataset configured for IIoT scenarios, using three representative backbones: AllCNN [41], UnResNet9, and LeNet-32 [42].

A. Experimental Settings

All experiments were run on a single NVIDIA RTX 4060 GPU (8GB). The base model was trained for 25 epochs with a batch size of 256. We evaluate two setups: Y1 forgets a single class (unknown), and Y2 forgets multiple classes (unknown, class 4, and class 5). These two configurations span

Algorithm 1 Multi-Teacher Knowledge Distillation for Machine Unlearning

- 1: **Input:** Base model $\mathcal{M}$, retain data $\mathcal{D}_{\mathcal{R}}$, forget data $\mathcal{D}_{\mathcal{F}}$, zero-day generator $G(\mathbf{z})$;
- 2: **Output:** Unlearned student model S ;
- 3: **Phase 1: Data Preprocessing**
- 4: **for** $j \leftarrow 1$ to M **do**
- 5: Sample $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ or select strategy $s \in \{\text{Mutation}, \text{Adversarial}\}$;
- 6: Generate $\tilde{\mathbf{f}}_j$ via Eq. (5), (6), or (7) with label y_{unknown} ;
- 7: Add $(\tilde{\mathbf{f}}_j, y_{\text{unknown}})$ to dataset $\mathcal{D}_{\text{aug}}$ via Eq. (9);
- 8: **end for**
- 9: **Phase 2: Unlearning Model Construction**
- 10: Initialize T_1 and T_2 with $\mathcal{M}$;
- 11: Optimize noise tensor $\mathbf{N}^*$ on $\mathcal{D}_{\mathcal{F}}$ via Eq. (14);
- 12: Update T_1 using $\mathcal{D}_{\mathcal{R}} \cup (\mathcal{D}_{\mathcal{F}} + \mathbf{N}^*)$;
- 13: Update T_2 on $\mathcal{D}_{\mathcal{R}}$ via Eq. (15);
- 14: **Phase 3: Student Model Generation and Training**
- 15: Initialize Student S ;
- 16: **for** batch $(\mathbf{x}, y)$ in $\mathcal{D}_{\mathcal{R}}$ **do**
- 17: Calculate logits $f_S(\mathbf{x}), f_{T_1}(\mathbf{x}), f_{T_2}(\mathbf{x})$;
- 18: Compute KL divergence losses $\mathcal{L}_{\text{KD}}^{(T_1)}$ and $\mathcal{L}_{\text{KD}}^{(T_2)}$ via Eq. (17) with temperature τ ;
- 19: Compute combined student loss $\mathcal{L}_{\text{student}}$ via Eq. (18);
- 20: Update S by minimizing $\mathcal{L}_{\text{student}}$;
- 21: **end for**
- 22: **return** S ;

single-class to multi-class forgetting regimes, thereby providing comprehensive coverage of realistic partial unlearning scenarios. Specifically, Y1 validates the framework’s capability to precisely forget a single attack class while preserving detection performance on all retained classes, representing a fundamental unlearning requirement. Y2 extends this to multi-class unlearning, requiring the model to simultaneously forget knowledge of multiple classes while maintaining robust performance on the remaining attack categories, thereby reflecting more complex real-world scenarios where multiple threats must be collectively removed. Together, these complementary settings enable thorough evaluation of the MT-KD framework across different forgetting scales and validate its effectiveness in both single-threat and multi-threat unlearning scenarios commonly encountered in practical IIoT systems.

To better reflect real IIoT conditions where attack patterns can shift and unknown threats appear, and to test robustness to zero-day and unseen attacks, we synthesize three types of random attacks: mutations of known attacks, GAN-based samples, and randomly generated abnormal feature vectors. The perturbation magnitude η is set to 0.1–0.3 to keep variability within a realistic range.

To ensure fair comparison and facilitate reproducibility, we specify the key simulation parameters as follows: all three models are trained for 40 epochs using the Adam optimizer with learning rates of 0.01 for UnResNet9 and 0.1 for LeNet32 and AllCNN. Knowledge distillation parameters include temperature parameter $\tau = 4.0$, balance weight $\alpha = 0.7$, and noise regularization coefficient $\lambda = 0.1$.

During the training of the original model, we used the Adam optimizer with an initial learning rate of 0.01. When validation performance plateaued, the learning rate was reduced to help convergence. For the noise-based forgetting model, we kept the perturbation size moderate: the noise tensor $\mathbf{N}$ was clipped to $[0.1, 0.3]$. This choice kept the retained set $\mathcal{D}_{\mathcal{R}}$ largely unchanged while reliably disturbing the forget set $\mathcal{D}_{\mathcal{F}}$.

B. Evaluation Metrics

In prior studies, many metrics have been used to judge the effectiveness of *machine unlearning*. A single overall classification accuracy, however, is not suitable here because it does not match the goal of forgetting. During unlearning, the model is expected to lose the ability to recognize the forgotten classes, so their accuracy will drop by design. If we only look at overall accuracy, this drop can be hidden, and the real effect of unlearning may be obscured.

To address this, we use two metrics: Dr and Df . Dr is the accuracy on the retained classes. These are the classes the model should remember. Df is the accuracy on the forgotten classes. These are the classes the model should not recognize.

An ideal unlearning process drives the Df accuracy toward zero, meaning the model no longer recognizes the forgotten classes and instead misclassifies them. At the same time, the Dr accuracy should stay close to that of the original model, so performance on the retained classes is not harmed. This balance between forgetting and retention is key to judging the effectiveness and safety of unlearning methods.

Additionally, to evaluate the privacy protection capability of our unlearning approach, we employ Membership Inference Attack (MIA) as a privacy-oriented metric. MIA attempts to determine whether specific data samples were used in the model's training process. In the context of machine unlearning, a successful unlearning method should make it difficult for an attacker to infer the participation of forgotten data in the original training. We measure the attack success rate of MIA on both the original model and the unlearned model. A significant reduction in attack precision after unlearning indicates that the influence of the target data has been effectively removed, thereby enhancing privacy protection. Lower MIA success rates on the unlearned model demonstrate more complete erasure of the forgotten data's traces.

C. Data Processing

In this work, we use CIC-IDS2017 and USTC-TFC2016 as the main source of raw network traffic and process it with the USTC-TK2016 toolkit. We build a preprocessing pipeline to align the outputs with the CIC-IDS2017 classification scheme. First, we split the original PCAP files into flows using a slicing tool, turning long traces into small, manageable segments to ease feature extraction and later transformation. Then, we remap and standardize the flow labels to the ten classes defined by CIC-IDS2017, ensuring label consistency and compatibility across datasets.

To use image-based deep learning for IIoT intrusion detection, we convert each flow segment into a 28×28 grayscale image through a mapping that encodes byte sequences and

TABLE I
UNLEARNING RESULTS ON CIC-IDS2017 DATASET

Y	Model	Metric	Original Model	T_1	T_2	Student Model	EMN
Y1	ALLCNN	Dr	0.97	0.94	0.86	0.91	0.87
		Df	0.98	0	0	0	0
	UnResNet9	Dr	0.97	0.94	0.90	0.95	0.93
		Df	0.99	0	0	0	0
	LeNet32	Dr	0.94	0.91	0.82	0.88	0.90
		Df	0.89	0	0	0	0
Y2	ALLCNN	Dr	0.96	0.92	0.92	0.97	0.96
		Df	0.97	0	0	0	0
	UnResNet9	Dr	0.94	0.97	0.96	0.96	0.92
		Df	0.98	0	0	0	0
	LeNet32	Dr	0.96	0.96	0.92	0.93	0.91
		Df	0.89	0	0	0	0

TABLE II
UNLEARNING RESULTS ON USTC-TFC2016 DATASET

Y	Model	Metric	Original Model	T_1	T_2	Student Model	EMN
Y1	ALLCNN	Dr	0.89	0.87	0.91	0.92	0.88
		Df	0.90	0	0	0	0
	UnResNet9	Dr	0.92	0.93	0.90	0.91	0.90
		Df	0.93	0	0	0	0
	LeNet32	Dr	0.88	0.89	0.85	0.89	0.88
		Df	0.89	0	0	0	0
Y2	ALLCNN	Dr	0.87	0.87	0.89	0.90	0.88
		Df	0.88	0	0	0	0
	UnResNet9	Dr	0.94	0.92	0.95	0.96	0.91
		Df	0.93	0	0	0	0
	LeNet32	Dr	0.92	0.93	0.92	0.91	0.87
		Df	0.90	0	0	0	0

packet-level attributes such as payload bytes, packet sizes, and inter-arrival times. This image representation is particularly suitable for IIoT environments where industrial protocols exhibit periodic communication patterns and structured byte sequences. Normal IIoT traffic manifests as recognizable visual patterns and textures in images, while attacks that disrupt protocol structure appear as irregular pixel distributions and visual anomalies. This encoding allows CNNs to effectively learn and distinguish IIoT-specific normal and malicious communication patterns.

Further, we pad or crop each sample to a fixed 28×28 shape and apply per-image normalization to stabilize training. To improve generalization, we apply standard augmentations such as random rotation, horizontal flip, light random crops, and brightness and contrast jitter. Finally, we create stratified train, validation, and test splits that preserve class ratios and reduce leakage by keeping flows from the same session in the same partition.

D. Unlearning Performance Evaluation

We conducted comprehensive experiments to test the multi-teacher knowledge distillation unlearning method on multiple datasets, backbones, and forgetting settings. Tables I and II report results on CIC-IDS2017 and USTC-TFC2016 for single-class (Y1) and multi-class (Y2) forgetting.

From Tables I and II, the baseline models perform well, with CIC-IDS2017 reaching 0.94–0.99 accuracy and USTC-TFC2016 reaching 0.87–0.94. Under the T_1 setting, Dr usually changes by only 3–5% while Df reaches 0. For

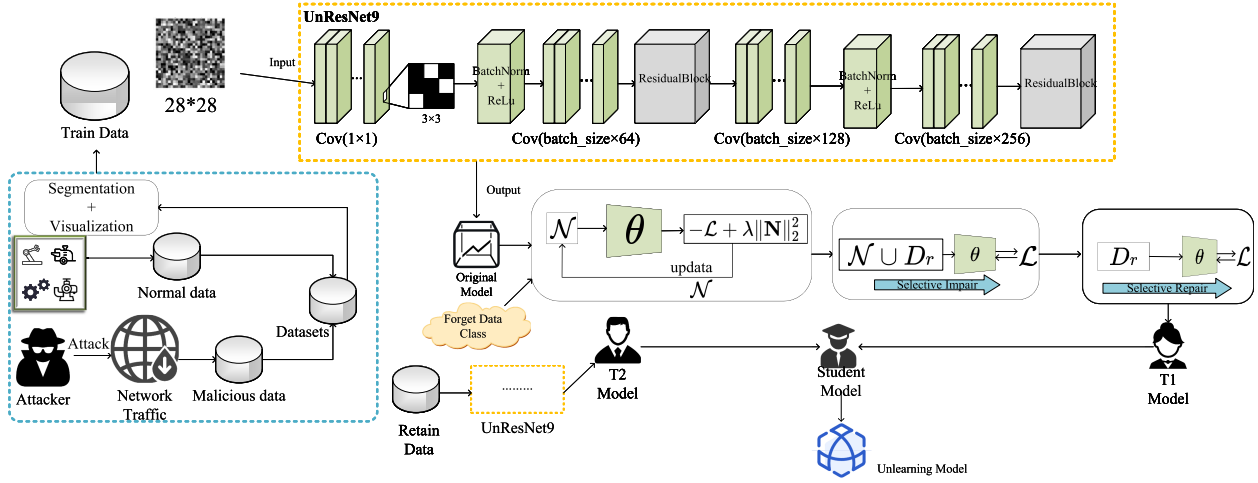


Fig. 2. Detailed architecture of the MT-KD unlearning framework for IIoT intrusion detection scenarios.

example, in CIC-IDS2017 with Y1 on UnResNet9, D_r drops by just 0.03.

Although the T_1 model keeps accuracy above 87% in all settings, noise-only forgetting cannot fully remove the effect of the forgotten data. To address this, we use a multi-teacher distillation setup that pairs T_1 with T_2 , where T_2 is trained only on the retained set. This yields consistent accuracy gains on both datasets. On CIC-IDS2017 (Y2-ALLCNN), the student reaches 0.97, up from 0.96 for the original. On USTC-TFC2016, Y2-UnResNet9 reaches 0.96, up from 0.94. These results show that multi-teacher distillation reduces side effects from noise-based training and keeps performance on the retained classes high during forgetting.

Compared with error-maximizing noise (EMN) [26], our method delivers consistently better results across all tested settings. On both CIC-IDS2017 and USTC-TFC2016, the student improves by roughly a few percentage points, with the most noticeable gains in the single-class forgetting setup and clear advantages preserved in multi-class cases. This trend is stable across the different backbones we evaluated.

Compared with EMN unlearning, our method needs slightly more training time—about 10–15% extra in our runs—but it consistently delivers higher accuracy on the retained classes. This small compute premium is offset by better detection, in particular by preserving key decision boundaries after forgetting. Across both datasets, the evidence shows that the time–accuracy trade-off clearly favors our approach. In practical IIoT deployments, where accuracy and robustness are critical to defend against evolving threats, this trade-off is reasonable and beneficial.

Fig. 4 illustrates the performance of the ALLCNN model under our proposed unlearning method. The original model, trained without any unlearning, already achieves excellent performance with an accuracy of 90%. After 20 epochs of training, both the T_1 and T_2 models show performance trends converging toward that of the original model. Furthermore, after 40 epochs of training, the student model achieves performance that is nearly identical to the original model.

Fig. 5 illustrates the trend of the D_r of the LeNet32 model over training epochs following the application of our proposed

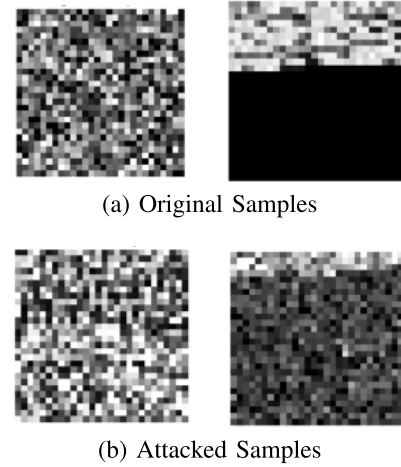


Fig. 3. (a) Images generated by preprocessing CIC-IDS2017; (b) fake samples generated by randomly attacking original samples to simulate zero-day attacks.

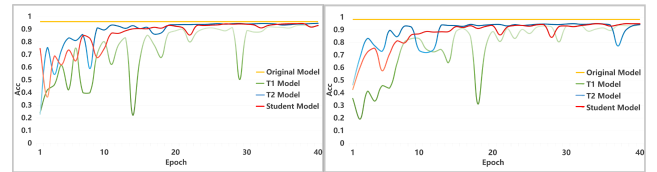


Fig. 4. The D_r performance of the ALLCNN model is evaluated under different attack intensities by setting the intensity to 0.1 (left) and 0.3 (right), respectively.

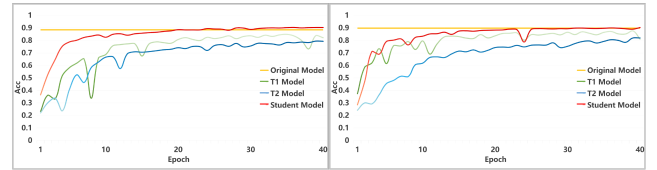


Fig. 5. The D_r performance of the LeNet32 model is evaluated under different attack intensities by setting the intensity to 0.1 (left) and 0.3 (right), respectively.

unlearning method. It is observed that the original model, without any unlearning intervention, consistently maintains a high and stable D_r , with accuracy close to 0.9; however, it

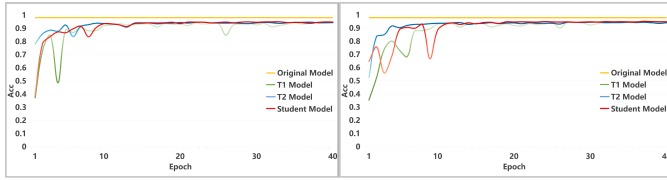


Fig. 6. The D_r performance of the UnResNet9 model is evaluated under different attack intensities by setting the intensity to 0.1 (left) and 0.3 (right), respectively.

performs approximately 9% worse in accuracy compared to the ALLCNN model. During training, the performances of the T_1 and T_2 models gradually improve, approaching that of the original model after around 20 epochs. By about 40 epochs, the student's D_r is close to that of the original model. At a higher attack level of 0.3, performance dips early but recovers over training and ends near the baseline. In our runs, the curves stay stable across the tested intensities.

Fig. 6 plots the D_r trajectory for UnResNet9. Compared with the other two backbones, its curve is smoother and more stable, indicating a steadier optimization process and better convergence. During the base training phase, UnResNet9 improves faster and reaches high accuracy in fewer epochs, which points to strong capacity and good generalization. It also achieves the best accuracy overall, yielding more reliable detection. Taken together—higher accuracy and quicker convergence—UnResNet9 is a strong choice for our unlearning framework. It adapts well to the forgetting steps and remains robust under different attack intensities, consistent with our objectives.

While the original models scored above 90% on D_f , our unlearning reduces this metric to 0. Across all three backbones, the student models consistently forget the target classes. This consistent drop of D_f to zero supports the method's effectiveness and its ability to generalize across architectures.

E. Prediction Distribution for the Forgotten Class

Fig. 7 compares predictions before and after applying our unlearning framework. In Fig. 7a, the original ALLCNN predicts all classes well, including the class to be forgotten. After Y1 unlearning, the student in Fig. 7c assigns almost no confidence to the forgotten class, and we do not observe confusion with other labels, which shows clean class-level forgetting. Fig. 7d reports results for UnResNet9 under Y2, which forgets classes 4, 5, and unknown; the student no longer maps any samples from these classes to retained labels. This pattern indicates that the model has removed not only the output responses but also the internal feature representations linked to the forgotten classes.

In conclusion, the proposed framework reaches a practical balance between targeted forgetting and preservation of utility. Across datasets and backbones, it consistently removes class-specific attack knowledge, driving D_f to zero, while maintaining high recognition on benign traffic and retained attack types. The strength of erasure is controllable through the forgetting and distillation weights, which allows tuning the trade-off to deployment needs. These results indicate

TABLE III
ATTACK PRECISION OF MIA ON UNLEARNED MODELS

Dataset	Original	T_1	T_2	Student
CIC-IDS2017	89.3	52.2	48.8	46.9
USTC-TFC2016	88.6	54.5	50.7	49.4

robustness and real-world value for IIoT intrusion detection, where models must adapt to evolving threats without full retraining.

F. Privacy Protection Evaluation via Membership Inference Attacks

Table III reports MIA precision [43] before and after unlearning. The original model is highly exposed, with MIA precision at 89.3% on CIC-IDS2017 and 88.6% on USTC-TFC2016, indicating strong privacy leakage. After applying our method, the T_1 model reduces attack precision to 52.2% and 54.5%. Training only on T_2 lowers it further to 48.8% and 50.7%, close to the 50% random-guess level. The final Student model offers the best protection, reaching 46.9% and 49.4%, a relative improvement of 47.5% and 44.2% over the original model. At this near-random level, an adversary cannot reliably distinguish training records, supporting the privacy gains of our framework.

To evaluate privacy during forgetting, we use two metrics: Forgetting AUC and Retention AUC. Both are computed from the ROC curve of an MIA. Forgetting AUC measures how well an attacker can identify forgotten samples; when it is close to 0.5, the attacker is no better than random guessing, indicating good privacy. Retention AUC measures the privacy of retained samples; values at 0.5 or below show that the attack has been disrupted or even misled, offering stronger protection than simple data deletion.

Figs. 8 and 9 present the comprehensive evaluation of the proposed unlearning framework across three neural network architectures on CIC-IDS2017 and USTC-TFC2016 datasets, respectively. The experimental results reveal distinct dataset-dependent characteristics while validating the universal effectiveness of the unlearning methodology.

Fig. 8 shows that all three backbones push the Forgetting AUC below 0.5 on CIC-IDS2017, which makes membership inference close to random. ALLCNN reaches about 0.48 within 15–20 epochs (roughly 4% under the 0.5 line) while keeping 85–90% validation accuracy. UnResNet9 is similar, settling near 0.49 in the same epoch range with comparable accuracy. The smaller LeNet32 takes a bit longer (20–25 epochs) but ends at 0.48–0.49 and holds 80–85% accuracy. These curves show steady unlearning with stable accuracy for all three models on this dataset.

Fig. 9 shows that all three architectures converge to a very similar forget AUC of about 0.49 on the USTC-TFC2016 dataset. While the final privacy level is comparable, the path to reach it shows clear trade-offs in speed and retained accuracy. UnResNet9 in Fig. 9b is the fastest, reaching the target in 15–20 epochs and keeping validation accuracy at 90–95%. ALLCNN in Fig. 9a needs 20–25 epochs and holds 85–90%

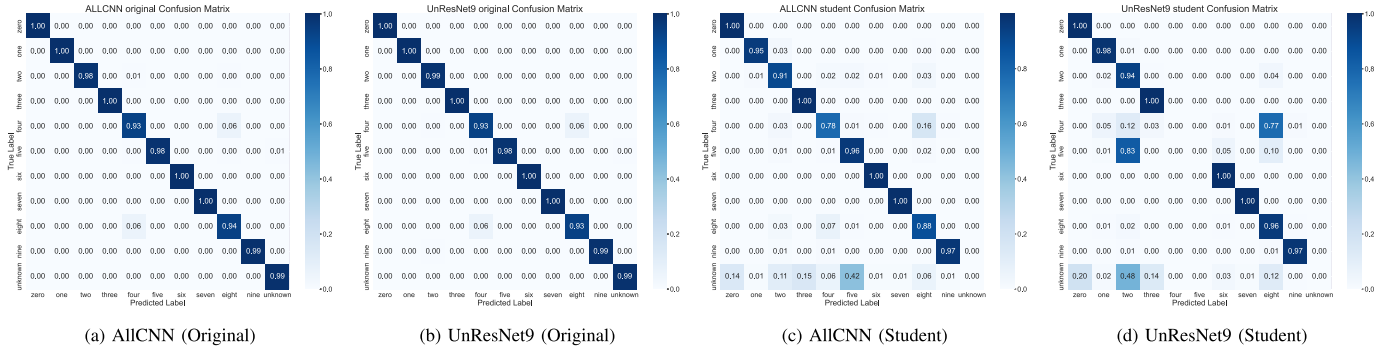


Fig. 7. Comparison of the confusion matrices of UnResNet9 and AIIcNN after applying the unlearning method.

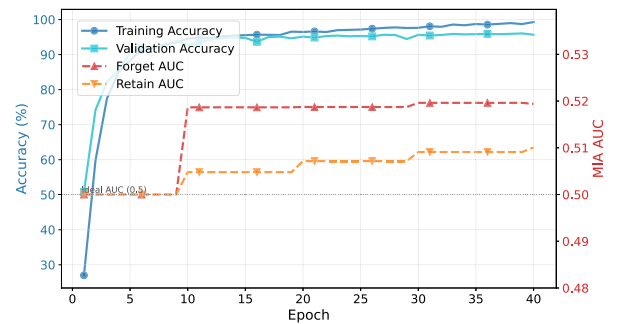
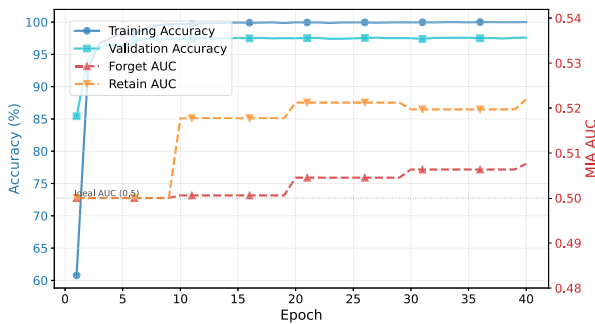
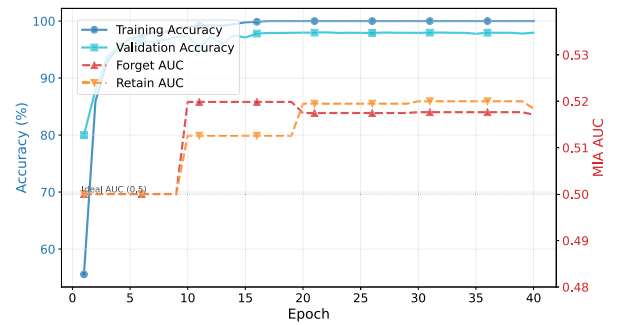
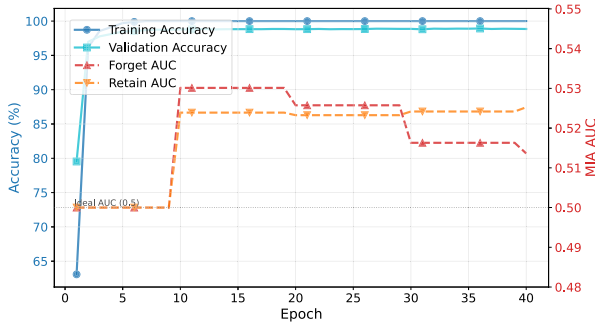
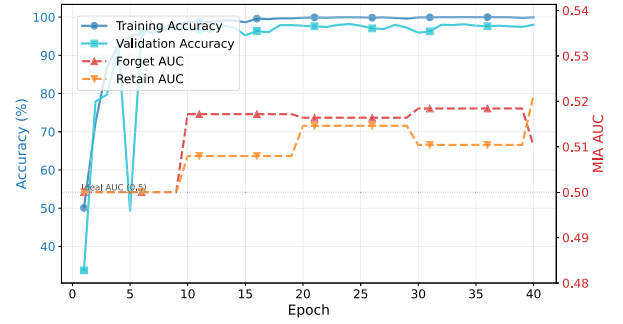
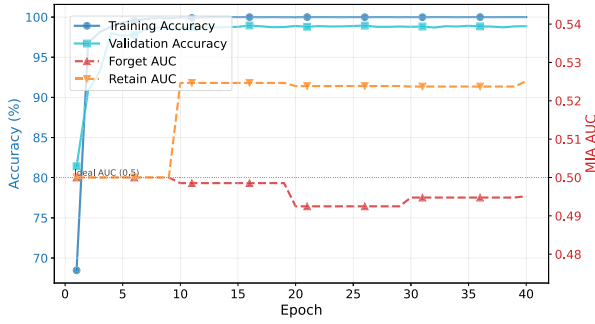


Fig. 8. AUC comparison on CIC-IDS2017 dataset across three architectures.

Fig. 9. AUC comparison on USTC-TFC2016 dataset across three architectures.

accuracy. LeNet32 in Fig. 9c takes 25–30 epochs and reaches 80–85% accuracy. This shared forget AUC suggests the traffic in USTC-TFC2016 sets a similar privacy floor for all three

models, while stronger backbones like UnResNet9 still offer faster convergence and higher accuracy. In practice, this means UnResNet9 is a good choice when time and accuracy matter

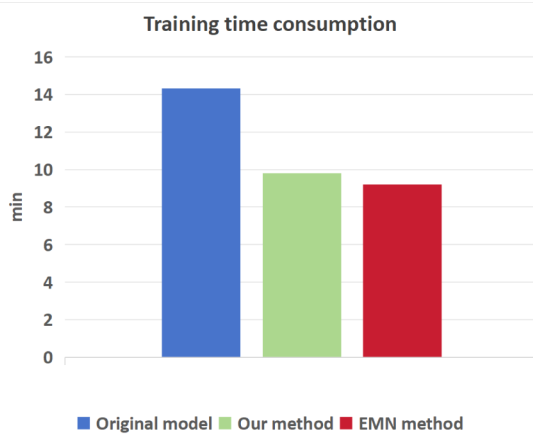


Fig. 10. Training time comparison.

most, whereas lighter models remain useful when compute or memory is limited.

Comparing Figs. 8 and 9 shows the same pattern: in all tested settings, the forget AUC reaches the 0.5 random-guess level or lower. Hitting this at-random mark means membership inference attacks no longer work in practice, giving stronger privacy than simple record deletion. The plots also suggest a link between model size and unlearning speed: UnResNet9 reaches the target fastest on both datasets, achieving it in just 15 to 20 epochs, followed by AllCNN at 15 to 25 epochs, with LeNet32 taking the longest at 20 to 30 epochs. This ordering holds across both Y1 and Y2 and is consistent with the accuracy trends reported above.

Across both figure sets, the retain AUC stays stable near the random-guess level of 0.51–0.52 in all settings. This shows that the unlearning step focuses on the forgotten data without hurting the privacy of retained samples. At the same time, model utility remains strong: validation accuracy is high, with UnResNet9 at 85–95%, AllCNN at 85–90%, and LeNet32 at 80–85%. Keeping retain AUC near random while preserving accuracy supports the practical value of our framework for IIoT intrusion detection.

As shown in Figs. 8 and 9, the framework consistently drives the forget AUC to the random-guess range (about 0.5) or lower across datasets and architectures. Combined with the preserved high validation accuracy, this supports its practical use for privacy-preserving intrusion detection. We also observe a clear robustness pattern: on USTC-TFC2016, all backbones converge to a similar forget AUC, while on CIC-IDS2017 the outcome depends on the architecture (with faster convergence for UnResNet9), indicating that the method adapts to different traffic characteristics without sacrificing utility.

G. Training Time Comparison

Fig. 10 presents an overall comparison of the training times for the different methods. The original model requires a longer training time because it must be trained from scratch without any prior knowledge. Compared to the EMN forgetting learning method, the method proposed in this study appropriately limits the increase in training time. This moderate cost is

justified by the significant improvement in model accuracy. In contrast, the EMN forgetting learning method achieves shorter training time, but the improvement in prediction performance is limited. Overall, this method offers an excellent balance between computational efficiency and accuracy improvement, and has outstanding practical value for industrial intrusion detection systems in the Internet of Things, both in terms of training cost and detection performance.

H. Sensitivity Analysis of Key Parameter α

To scientifically validate the optimal parameter α , we conducted a sensitivity analysis on CIC-IDS2017 using UnResNet9, evaluating the MT-KD framework across four α values: {0.6, 0.7, 0.8, 0.9}. The results reveal that D_r exhibits a clear optimum at $\alpha = 0.7$ with 95.50% accuracy, outperforming $\alpha = 0.6$, $\alpha = 0.8$, and $\alpha = 0.9$ with 94.4%, 94.9%, and 93.6%, respectively. This non-monotonic relationship demonstrates that $\alpha = 0.7$ represents a carefully optimized choice rather than an arbitrary selection. Critically, all α values achieve 0.00% accuracy on D_f , confirming that the dual-teacher distillation mechanism effectively enforces forgetting regardless of parameter weighting. The superior retention performance combined with complete forgetting at $\alpha = 0.7$ validates that our parameter selection is data-driven and theoretically justified: allocating 70% influence to the forget-focused teacher T_1 and 30% to the retain-focused teacher T_2 achieves optimal balance between knowledge preservation and effective unlearning.

I. Discussion

MT-KD framework employs a dual-teacher knowledge distillation architecture. The dual-teacher configuration demonstrates strong effectiveness for multi-class unlearning in IIoT systems: the T_1 provides explicit forgetting guidance through noise optimization, while the T_2 ensures robust performance preservation on retained classes. Experimental results confirm this effectiveness—the MT-KD framework achieves near-zero accuracy on forget classes while maintaining high performance on retained classes. The dataset used in the current experiments is relatively limited in scale, and the dual-teacher configuration is sufficient to meet experimental requirements. Extensive evaluations across various neural network architectures demonstrate that the dual-teacher distillation strategy’s capability to prevent information leakage is robust and generalizable.

However, as dataset scales expand and real-world deployment scenarios become more complex, multi-teacher frameworks are expected to demonstrate increasingly significant advantages in computational efficiency and unlearning performance. Additional teachers could provide complementary perspectives on forgetting and retention, potentially improving unlearning precision. The computational implications are important: while current experiments show efficient training without operational overhead, increasing the number of teachers would proportionally increase computational cost, which must be carefully balanced against benefits, particularly

for resource-constrained IIoT edge devices. Therefore, extending the dual-teacher framework to multi-teacher configurations represents a promising direction for future research, where increased model complexity and dataset scales may unlock additional benefits while maintaining alignment with privacy regulations and the right to be forgotten.

VI. CONCLUSION

We have proposed an MT-KD framework for privacy-aware machine unlearning in IIoT intrusion detection systems. By converting network traffic into grayscale images and providing a theoretical framework aligned with privacy regulations including the right to be forgotten, our method addresses critical challenges in stricter IIoT intrusion detection requirements. The dual-teacher architecture achieves complementary guidance through noise optimization and data preservation, demonstrating strong effectiveness in multi-class unlearning tasks specific to IIoT. Training time comparisons confirm efficient removal of targeted data influence without degrading operational efficiency, making the method practical for edge devices. Extensive evaluations across various neural network architectures demonstrate its capability to prevent information leakage through rigorous privacy evaluation via membership inference attacks. This work opens new directions for machine unlearning in rigorous IIoT intrusion detection scenarios.

Future work will follow a more structured and systematic approach with concrete objectives. We plan to extend MT-KD to deeper network architectures including ResNet50/101, DenseNet121. Beyond empirical evaluation, we will establish rigorous quantitative metrics to measure unlearning efficacy and assess potential information leakage through membership inference attacks and model extraction attacks. We will also conduct critical analysis of the method's limitations, including performance under extreme forgetting scenarios and computational cost-accuracy trade-offs. These efforts will provide verifiable guarantees on unlearning effectiveness and establish theoretical bounds on residual model information in large-scale IIoT environments.

REFERENCES

- [1] J.Sun et al., "Privacy-preserving fine-grained data sharing with dynamic service for the cloud-edge IoT," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 2, pp. 1329–1346, Mar.2025.
- [2] S.Shen, L.Xie, Y.Zhang, G.Wu, H.Zhang, and S.Yu, "Joint differential game and double deep Q-networks for suppressing malware spread in industrial Internet of Things," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 5302–5315, Aug. 2023.
- [3] X.Yang, W.Wang, T.Tian, and C.Wang, "Cryptanalysis and improvement of a blockchain-based certificateless signature for IIoT devices," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 1884–1894, Feb.2024.
- [4] S.Shen, C.Cai, Y.Shen, X.Wu, W.Ke, and S.Yu, "Joint mean-field game and multiagent asynchronous advantage actor-critic for edge intelligence-based IoT malware propagation defense," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 4, pp. 3824–3838, Jul.2025.
- [5] Y.Shen, C.Shepherd, C. M.Ahmed, S.Shen, and S.Yu, "SGD3QN: Joint stochastic games and dueling double deep Q-networks for defending malware propagation in edge intelligence-enabled Internet of Things," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 6978–6990, Jun. 2024.
- [6] L.Cui, Y.Li, Y.Zhou, Y.Qu, and J.Liu, "Accelerating blockchain-enabled federated learning with clustered clients," *IEEE Trans. Big Data*, vol. 11, no. 5, pp. 2148–2161, Oct.2025.
- [7] C.Sheng et al., "Network traffic fingerprinting for IIoT device identification: A survey," *IEEE Trans. Ind. Informat.*, vol. 21, no. 5, pp. 3541–3554, May.2025.
- [8] M.Sharma, A.Tomar, and A.Hazra, "Edge computing for industry 5.0: Fundamental, applications, and research challenges," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19070–19093, Jun.2024.
- [9] J.Wang et al., "A zero-shot federated unlearning framework with stability verification," *IEEE Trans. Cognit. Commun. Netw.*, vol. 12, pp. 2021–2035, Dec. 2025.
- [10] Y.Bao, J.Sun, X.Cheng, W.Qiu, and L.Nie, "Blockchain-based privacy-preserving alternative credit data sharing," *IEEE Trans. Computat. Social Syst.*, vol. 13, no. 1, pp. 1–15, Feb.2026.
- [11] Y.Qu, X.Yuan, M.Ding, W.Ni, T.Rakotoarivelo, and D.Smith, "Learn to unlearn: Insights into machine unlearning," *Computer*, vol. 57, no. 3, pp. 79–90, Mar.2024.
- [12] J.Sun et al., "A tamper-resistant broadcasting scheme for secure communication in Internet of Autonomous Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 2837–2846, Mar.2024.
- [13] Z.Ran, Y.Yao, W.Li, W.Yang, W.Li, and Y.Wu, "Precise defense approach against small-scale backdoor attacks in industrial Internet of Things," *IEEE Internet Things J.*, vol. 12, no. 5, pp. 5742–5754, Mar.2025.
- [14] Y.Lin et al., "Decentralized unlearning for trustworthy AI-generated content (AIGC) services," *IEEE Netw.*, vol. 39, no. 3, pp. 285–292, May.2025.
- [15] H.Kheddar, D. W.Dawoud, A. I.Awad, Y.Himeur, and M. K.Khan, "Reinforcement-learning-based intrusion detection in communication networks: A review," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 4, pp. 2420–2469, Aug.2025.
- [16] M. M.Fouda, Z. M.Fadlullah, M. I.Ibrahim, and N.Kato, "Privacy-preserving data-driven learning models for emerging communication networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 4, pp. 2505–2542, Aug.2025.
- [17] L.Porcelli, U.Fiore, and F.Palmieri, "Generative models with helical time encoding for seasonal time series forecasting," *Eng. Appl. Artif. Intell.*, vol. 162, Dec.2025, Art. 112780.
- [18] N.Romandini, A.Mora, C.Mazzocca, R.Montanari, and P.Bellavista, "Federated unlearning: A survey on methods, design guidelines, and evaluation metrics," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 11697–11717, Jul.2025.
- [19] X.Wang, J.Zhang, C.Chen, J.He, Y.Ma, and X.Guan, "Trust-AoI-aware codesign of scheduling and control for edge-enabled IIoT systems," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 2833–2842, Feb.2024.
- [20] X.Li, C.Liu, S.Sun, Z.Zhou, Y.Zhang, and G.Li, "An efficient two-stage machine unlearning framework for poisoned specific emitter identification," *IEEE Internet Things J.*, vol. 12, no. 17, pp. 36662–36673, Sep.2025.
- [21] J.Zhang, M.Zhao, Z.Wang, W.Su, and P.Wang, "Model recovery in federated unlearning with restricted server data resources," *IEEE Internet Things J.*, vol. 12, no. 11, pp. 17920–17935, Jun.2025.
- [22] A.Zhang, P.Zhao, W.Lu, Y.Zhou, W.Zhang, and G.Zhang, "Mitigating poisoning attacks in federated learning through deep one-class classification," *IEEE Trans. Cognit. Commun. Netw.*, vol. 12, pp. 545–558, Dec.2025.
- [23] K.Xu et al., "Self-supervised learning malware traffic classification based on masked autoencoder," *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17330–17340, May.2024.
- [24] H.Hu, S.Wang, T.Dong, and M.Xue, "Learn what you want to unlearn: Unlearning inversion attacks against machine unlearning," in *Proc. IEEE Symp. Secur. Privacy (SP)*, San Francisco, CA, USA, May 2024, pp. 3257–3275.
- [25] H.Sun, T.Zhu, W.Chang, and W.Zhou, "Generative adversarial networks unlearning," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 5, pp. 5303–5320, Sep.2025.
- [26] A. K.Tarun, V. S.Chundawat, M.Mandal, and M.Kankanhalli, "Fast yet effective machine unlearning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 13046–13055, Sep.2024.
- [27] W.Wang, C.Zhang, Z.Tian, and S.Yu, "FedU: Federated unlearning via user-side influence approximation forgetting," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 3, pp. 2550–2562, May.2025.
- [28] V. S.Chundawat, A. K.Tarun, M.Mandal, and M.Kankanhalli, "Zero-shot machine unlearning," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2345–2354, Apr. 2023.
- [29] Y.Li, Z.Guo, N.Yang, H.Chen, D.Yuan, and W.Ding, "Threats and defenses in the federated learning life cycle: A comprehensive survey

and challenges,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 15643–15663, Sep.2025.

- [30] J.Tang et al., “Toward general industrial intelligence: A survey of large models as a service in industrial IoT,” *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 2054–2086, Jun.2026.
- [31] Y.Shan, Y.Yao, X.Zhou, T.Zhao, B.Hu, and L.Wang, “CFL-IDS: An effective clustered federated learning framework for industrial Internet of Things intrusion detection,” *IEEE Internet Things J.*, vol. 11, no. 6, pp. 10007–10019, Mar.2024.
- [32] Y.Hu et al., “Industrial Internet of Things intelligence empowering smart manufacturing: A literature review,” *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19143–19167, Jun.2024.
- [33] S.Yu et al., “Deep Q-network-based open-set intrusion detection solution for industrial Internet of Things,” *IEEE Internet Things J.*, vol. 11, no. 7, pp. 12536–12550, Apr.2024.
- [34] Q.Zhang, L.Tang, T.Chin-Hon, T.Shi, J.Xuan, and C.Yu-Chao, “GCVIF: Pioneering explainable domain-shared representation learning for fault signal detection in multiple working states simultaneously,” *IEEE Internet Things J.*, vol. 12, no. 3, pp. 2775–2789, Feb.2025.
- [35] G.Wu et al., “Combining Lyapunov optimization with actor-critic networks for privacy-aware IIoT computation offloading,” *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17437–17452, May.2024.
- [36] F.Zhang, G.Han, L.Liu, Y.Zhang, Y.Peng, and C.Li, “Cooperative partial task offloading and resource allocation for IIoT based on decentralized multiagent deep reinforcement learning,” *IEEE Internet Things J.*, vol. 11, no. 3, pp. 5526–5544, Feb.2024.
- [37] L.Wang, X.Zhang, H.Su, and J.Zhu, “A comprehensive survey of continual learning: Theory, method and application,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5362–5383, Aug.2024.
- [38] O.Aouedi et al., “A survey on intelligent Internet of Things: Applications, security, privacy, and future directions,” *IEEE Commun. Surveys Tuts.*, vol. 27, no. 2, pp. 1238–1292, Apr.2025.
- [39] X.Jiang, F. R.Yu, T.Song, and V. C. M.Leung, “Resource allocation of video streaming over vehicular networks: A survey, some research issues and challenges,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 5955–5975, Jul.2022.
- [40] M.Adil, H.Song, S.Mastorakis, H.Abulkasim, A.Farouk, and Z.Jin, “UAV-assisted IoT applications, cybersecurity threats, AI-enabled solutions, open challenges with future research directions,” *IEEE Trans. Intell. Vehicles*, vol. 9, no. 4, pp. 4583–4605, Apr.2024.
- [41] C.Mouand J.Zhang, “TransCL: Transformer makes strong and flexible compressive learning,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5236–5251, Apr.2023.
- [42] L.Luo, C.Zhang, H.Yu, G.Sun, S.Luo, and S.Dustdar, “Communication-efficient federated learning with adaptive aggregation for heterogeneous client-edge-cloud network,” *IEEE Trans. Services Comput.*, vol. 17, no. 6, pp. 3241–3255, Nov.2024.
- [43] H.Chen et al., “Practical membership inference attack against collaborative inference in industrial IoT,” *IEEE Trans. Ind. Informat.*, vol. 18, no. 1, pp. 477–487, Jan.2022.



Yizhou Shen received the B.Sc. degree (Hons.) in computer science from Cardiff University, Cardiff, U.K., in 2023. She is currently pursuing the Ph.D. degree in computer science with Newcastle University, Newcastle upon Tyne, U.K.

Her research interests include cyber security, game theory, deep reinforcement learning, edge intelligence, and the Internet of Things.



Jingnan Dong (Member, IEEE) received the Ph.D. degree in computer science and technology from Chongqing University of Posts and Telecommunications, Chongqing, China, in 2023. He is an Associate Professor with the School of Information Engineering, Huzhou University, Zhejiang, China. His research interests include blockchain technology, applied cryptography, network security, cloud computing, and federated learning.



Wenlong Ke received the Ph.D. degree in information and communication engineering from Guilin University of Electronic Technology, Guilin, China, in 2021. He is a Lecturer with the School of Information Engineering, Huzhou University, Huzhou, China. His current research interests include data-center networks, software-defined networks, and machine learning.



Tian Wang (Senior Member, IEEE) received the Ph.D. degree in computer science from the City University of Hong Kong in 2011. Currently, he is a Professor with the Institute of Artificial Intelligence and Future Networks, Beijing Normal University. He has 27 patents and has published more than 200 papers in high-level journals and conferences. He has more than 14 000 citations, according to Google Scholar. His H-index is 68. His research interests include the Internet of Things, edge computing, and mobile computing.



Shigen Shen (Senior Member, IEEE) received the Ph.D. degree in pattern recognition and intelligent systems from Donghua University, Shanghai, China, in 2013.

He is a Professor with the School of Information Engineering, Huzhou University, Huzhou, China. He has published more than 100 technical papers. His current research interests include the Internet of Things, cyber security, edge computing, and game theory.



Jiashuo Niu received the B.Eng. degree in the Internet of Things engineering in 2024. He is currently pursuing the M.S. degree in computer technology with the School of Information Engineering, Huzhou University, Huzhou, China. His research interests include machine unlearning, intrusion detection, and Industrial Internet of Things security.



Ruidong Li (Senior Member, IEEE) received the Ph.D. degree from the Department of Computer Science, University of Tsukuba, Tsukuba, Japan, in 2008. His research interests include big data, quantum networks, metaverse, future networks, and the IoT.

He was a recipient of the Best Paper Awards for IEEE ICC 2022 and IWCMC 2022 and the IEEE Network 2022 Editor Appreciation Award. He serves as the Vice Chair for the IEEE Internet Technical Committee (ITC). He served as the Chair for several

conferences, such as the General Chair for MOBILWare 2022, IEEE MSN 2021, and CPSCOM 2021; the Area TPC Chair for INFOCOM 2023; and the TPC Chair for ICNC 2024, IEEE MetaCom 2023, HotICN 2022, IWQoS 2021, and MSN 2020. He also serves on several prestigious editorial boards, including IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING, IEEE NETWORK, and IEEE INTERNET OF THINGS JOURNAL.